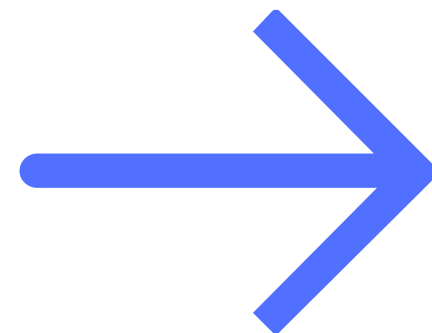


OBSERVABILITY IN

AI



HOW TO DEBUG RAG & AGENTS PROPERLY



TRACES 01

02 **METRICS**

EVALUATION
&
FEEDBACK 04

03 **LOGS**

THE
4
CORE
PILLARS

HRUTHIK GALI DILEEP

FOLLOW

↻ Repost

What is Observability?

Observability =

Your ability to look inside an AI system and understand what is happening.

Not just:

“The answer is wrong.”

But:

- Where did it fail?
- Retrieval?
- Prompt?
- Model?
- Tool call?

If you can't see inside, you're guessing.

Why It Matters in RAG

A RAG system is not just:

User → Retrieve → Generate

It's actually:

User

- Query rewrite
- Embedding
- Vector search
- Reranking
- Context building
- Prompt
- LLM

Failure can happen at ANY step.

Observability lets you trace it.

Observability in AI Agents

Agents are harder.

One user request can become:

User

- Agent reasoning
- Tool call
- Tool output
- Another tool
- Final answer

You must log:

- Each tool call
- Tool inputs & outputs
- Agent decisions

Otherwise debugging agents becomes chaos.

The 4 Pillars of AI Observability

- 1 Traces
- 2 Metrics
- 3 Logs
- 4 Feedback & Evaluation

These four together make your AI debuggable.

Let's break them down.

1 Traces (Full Story of One Request)

For every user query, you should see:

- Original query
- Rewritten query
- Retrieved chunks + scores
- Final context
- Prompt sent to LLM
- Model output

If the answer is wrong,
traces show you exactly where it broke.
This is the most important pillar for RAG.

2 Metrics (System Health)

Metrics tell you how your AI behaves at scale.

Track things like:

- Retrieval hit rate
- Similarity scores
- Token usage
- Latency per step
- Hallucination rate
- “No answer” rate

If similarity scores drop,
your knowledge base may be outdated.

Metrics = long-term visibility.

3 Logs (Engineering Layer)

Logs capture raw events:

- Vector DB timeout
- LLM failure
- Tool crash
- Empty response

Sometimes the issue isn't intelligence.
It's infrastructure.

Logs help you catch that.

4 Evaluation & Feedback(The Improvement Engine)

Evaluation & Feedback answer one big question:

Is the system actually getting better?

You need 3 layers here:

1 User Feedback

- 👍 / 👎 buttons
- “Was this helpful?”
- Report incorrect answers

Real users expose real failures.

2 Automated Evaluation

Use programmatic checks like:

- Faithfulness (Did the answer stick to context?)
- Relevance (Did it answer the question?)
- Context precision
- Groundedness

This helps you evaluate at scale without manual checking.

3 Benchmark / Test Sets

Create a fixed set of:

- Questions
- Expected answers
- Gold documents

Every time you change:

- Chunking
- Embeddings
- Prompt
- Model

Run the test set again.

If scores drop, you broke something.

What Happens Without Observability?

Without it:

You blame the LLM.

You tweak prompts randomly.

You over-engineer.

With observability:

You see:

- Wrong chunk retrieved
- Right chunk dropped
- Context too long
- Tool gave bad data

Now you fix the real issue.

Production AI is not about fancy prompts.

It's about:

Visibility.

Traceability.

Measurability.

If you cannot observe your AI,
you cannot improve it.

Build smart.

Observe everything.